

Assignment - D2

COD

PAGE NO.: 12

• Src - A (2 * 5 = 10)

1. Show the use of half modules of construct NAND and NOR gate with diagram.

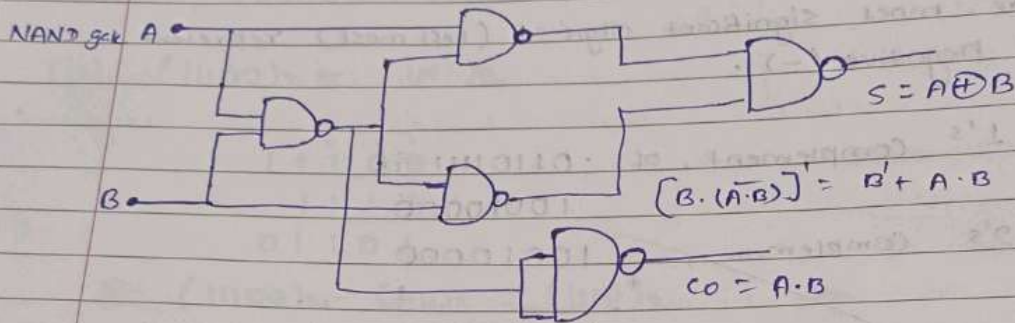
Ans

Half adder using NAND gate :-

$$S = A \oplus B$$

$$Co = A \cdot B$$

$$[A \cdot (\overline{A \cdot B})]' = A' + A \cdot B$$

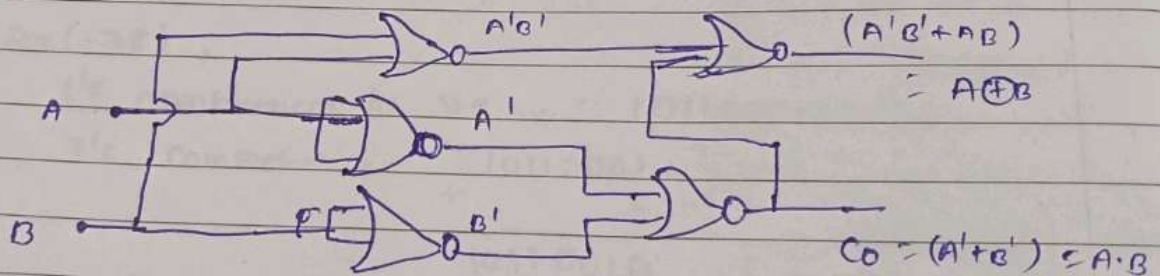


- Half adder using NOR gates :-

$$S = A \oplus B = A'B + B'A = (A'B' + AB)$$

$$C = A \cdot B$$

NOR gates



Q2.

What is decoder. Draw the logic diagram and Truth table.

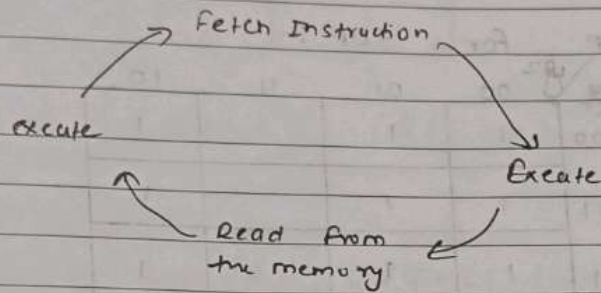
Ans

The combinational circuit that change the binary information into 2^N output lines is known as decoders.

There are many types of decoders:-

Q3. What is the purpose of fetch decode execute cycle?

Ans. The fetch-decode-execute cycle is the fundamental process in a CPU where it fetches instructions from memory, decodes them into operations, and executes those operations. Its purpose is to continuously fetch, interpret, and execute instructions to perform tasks and run programs on a computer.

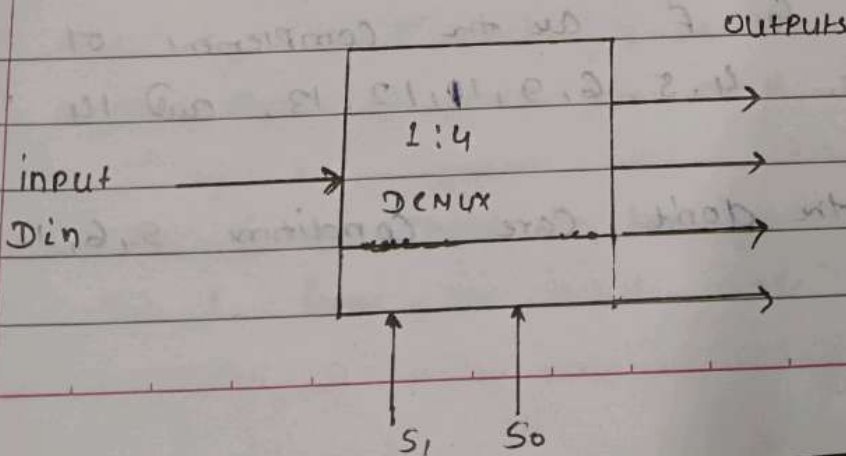


Q4. What components of the computer are involved in the fetch decode execute cycle?

Ans. The main components of a computer involved in the fetch-decode-execute cycle are the Central Processing Unit (CPU), specifically the control unit, arithmetic logic unit (ALU), and the registers. The memory unit is also significant involved.

Q5. Define the demultiplexer?

Ans. A demultiplexer is a digital circuit that takes a single input signal and chooses one of several output lines based on a set of control signals (also known as a demux or data selector).



Sec-B (3x7=21)

1. Simplify the Boolean function F together with the don't care conditions d in (1) SOP form and (2) POS form.

$$F(w, x, y, z) = \sum (0, 1, 2, 3, 7, 8, 10)$$

$$d(w, x, y, z) = \sum (5, 6, 11, 15)$$

Ans A. The given expression is
 $f(w, x, y, z) = \sum (0, 1, 2, 3, 7, 8, 10)$

K-map for SOP :-

wz	00	01	11	10
00	1	1		1
01	1	1		1
11	1	1		1
10				

SOP -

- (i) The minterms for F are 0, 1, 2, 3, 7, 8, 10.
 2. incorporating the don't care conditions 5, 6, 11, 15

∴ From the K-map we can see that

$$F(w, x, y, z) = \bar{w}y + x\bar{z} + yz$$

2. POS (Product of sums) :-

- (i) The maxterms for F are the complement of the minterms, 4, 5, 6, 9, 10, 12, 13, and 14.
 (ii) We include the don't care conditions 5, 6, 11 and 16.

K-map for POS:-

xy \ wz	00	01	11	10
00				
01				
11			1	
10	1	1	1	1

$$F(w, x, y, z) = (w + x + \bar{y}) \cdot (\bar{x} + \bar{y} + z) \cdot (\bar{x} + y + \bar{z})$$

So in POS form the simplified expression is -

$$F(w, x, y, z) = (w + x + \bar{y}) \cdot (\bar{x} + \bar{y} + z) \cdot (\bar{x} + y + \bar{z})$$

Q2. The content of a 4-bit register is initially 1101. The register is shifted six times to the right with the serial input being 101101. What is the content of the register after each shift?

Ans To solve this, let's consider each step.

- (i) Initially, the register content is 1101.
- (ii) After this first shift, the serial input 101101 enters from the left, so the new content becomes 1100 (1101 shifted right and 1 from the input added at the left).
- (iii) After the second shift, it becomes 0111 (1100 shifted right and 1 from the input added at the left).
- (iv) After the third shift, it becomes 1011 (0111 shifted right and 1 from the input added at the left).
- (v) After the fourth shift, it becomes 1101 (1011 shifted right and 1 from the input added at the left).

(vi) After the fifth shift, it becomes 1110 (1101 shifted right and 1 from the input added at the left).

(vii) After the sixth shift, it becomes 0111 (1110 shifted right and 1 from the input added at the left).

So the content of the register after each shift is:-

(i) 1110

(ii) 0111

(iii) 1011

(iv) 1101

(v) 1110

(v') 0111

Q3. How many 128×8 memory chips are needed to provide a memory capacity of 4096×16 ?

Ans Each 128×8 memory chips has a capacity of $128 \times 8 = 1024$ bits.

The total capacity required for a memory of $4096 \times 16 = 65536$ bits.

To find out how many chips are needed, we divide the total required capacity by the capacity of each chip.

$$\frac{65536 \text{ bits}}{1024 \text{ bits/chip}} = 64 \text{ chips}$$

Therefore, 64 chips are needed out of 1288 to provide a memory capacity of 4096×16 .

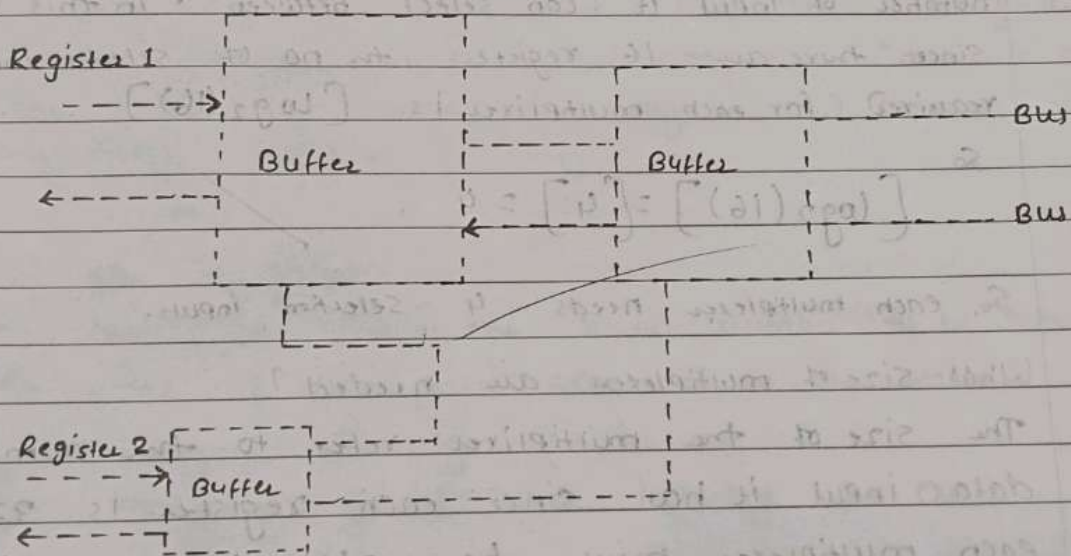
- Construct a bus system using tri state buffer for 2 registers having 4 bits each. Draw a neat and clean diagram to explain the process.

Ans

- Register :-** These are the storage units that hold binary data. In this case we have two registers, each with 4 bits.
- Tri-state Buffers :-** These are special logic gates that can be in one of three states, high(1), low(0) or high impedance (2).

In high impedance state, the buffer effect by disconnects the bus, allowing other devices to draw the bus without interference.

Diagram :-



in the diagram:

- There are 2 registers, each with 4 bits represented as Register 1 and Register 2.
- Each register is connected to a set of the state buffer. These buffers control the flow of data onto the bus.
- The bus is represented by a line connecting the two buffers.

4. Each buffer is controlled by a control signal.

5. When control signal high, the buffer allows the data from the register to pass onto the bus. When the control signal is low, the buffer disconnects the register from the bus, putting it in a high impedance state.

Q2. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers.

a. How many selection inputs are there in each multiplexer?

Ans. Each multiplexer has n selection inputs where n is the number of input it can select between. In this case, since there are 16 registers, the no. of selection input required for each multiplexer is $\lceil \log_2(16) \rceil$.

So,

$$\lceil \log_2(16) \rceil = \lceil 4 \rceil = 4$$

So, each multiplexer needs 4 selection inputs.

b. What size of multiplexers are needed?

Ans. The size of the multiplexer refers to the number of data input it has. Since each register is 32 bits, each multiplexer must be able to select between 32-bit data input. Therefore, the size of the multiplexer needed is 32.

c. How many multiplexers are there in the bus?

Ans. To determine the number of multiplexers needed in the bus, we need to consider the no. of bits in the register. Since each register is 32-bits

and there are 16 registers, there will be 16 data inputs to the multiplexer. Each multiplexer will have 16 data inputs and 4 selection inputs.

To handle all 32 bits of each register, we need to use 32 such multiplexers (one for each bit position). So, there will be 32 multiplexers in the bus.

Each multiplexer will select between 16 data inputs based on the 4 selection inputs.

Q3. Design a 4-bit combinational circuit incrementer and decrements using four full-adder circuits.

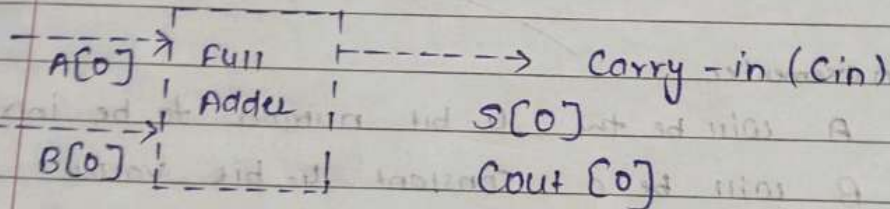
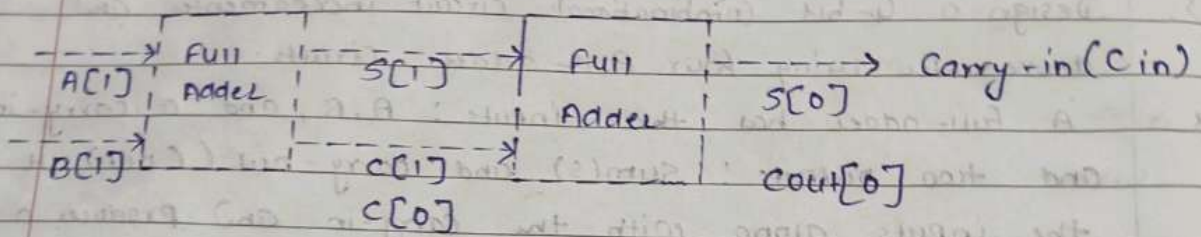
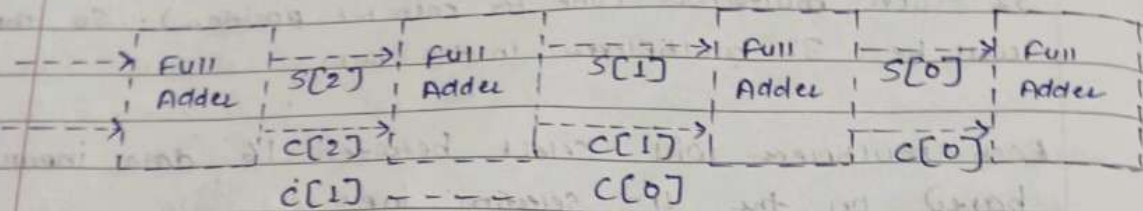
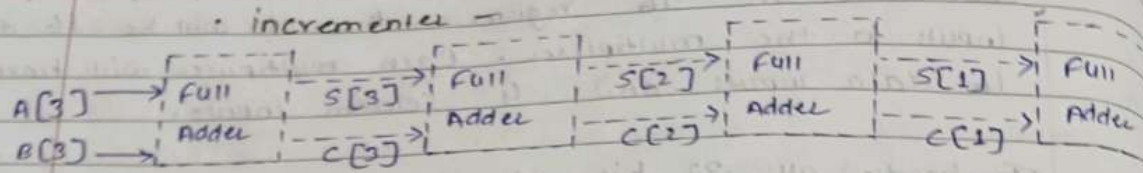
Ans. A full-adder has these inputs: A, B, and a carry-in (cin) and two outputs: Sum (S) and carry-out (Cout). It adds the inputs along with the carry-in and produces a sum and a carry-out.

For the incrementer:-

- The input A will be the 4-bit number to be incremented.
- The input B will be a constant 4-bit value representing 1 (binary 0001).
- The carry-in (cin) will initially be 0.

For the decrements:-

- The input A will be the 4-bit number to be decremented.
- The input B will be a constant 4-bit value representing -1 (binary 1111), considering 2's complement.
- The carry-in (cin) will initially be 1.

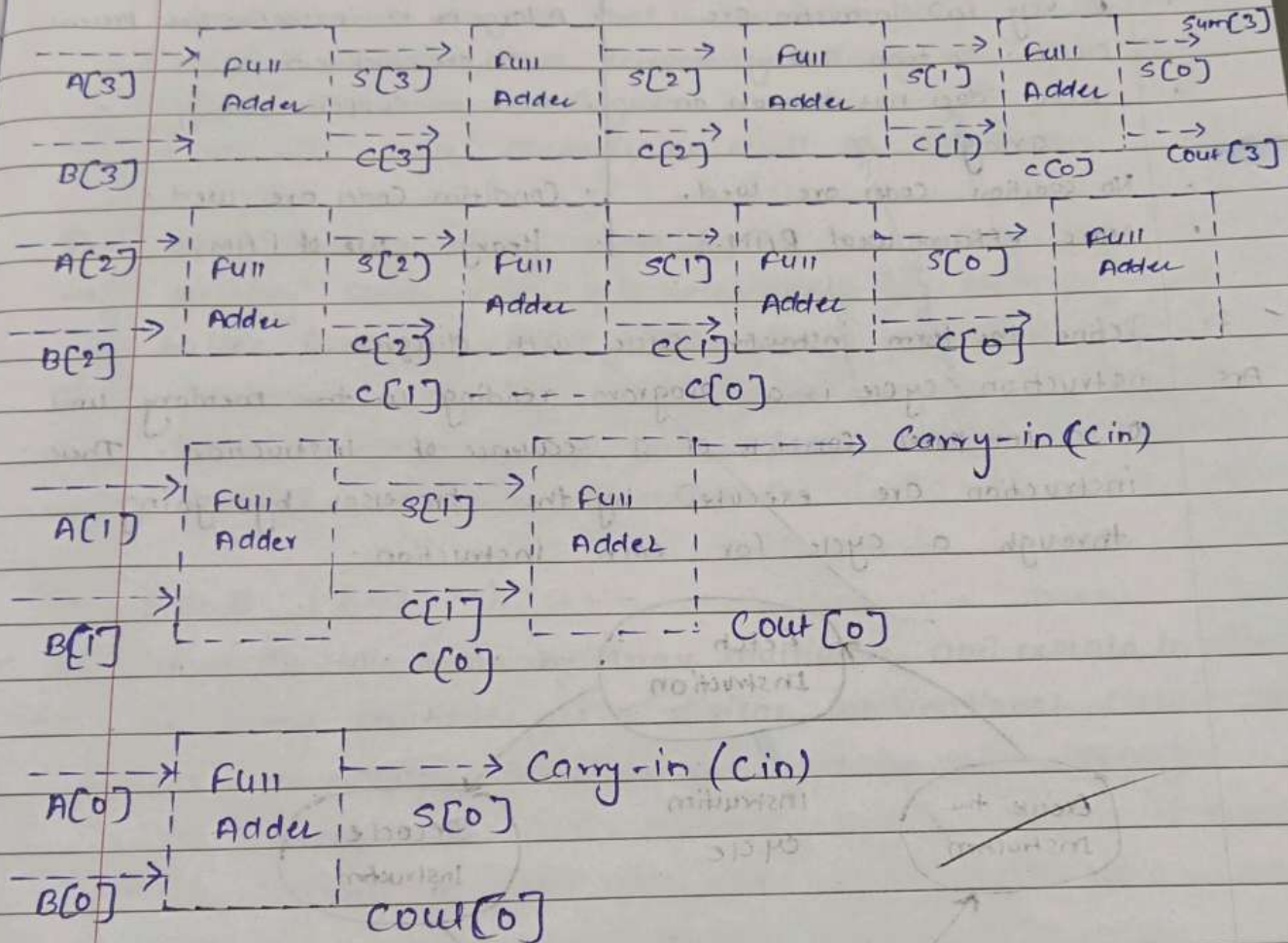


→ Sum[3]

S[0]

→ Cout[3]

• Decrementer :-



in both cases, the sum outputs represent the incremented or decremented values and the final Carry-out (Cout) represents overflow.

22BC011068